Evaluating functions

1

a x

b y

2 The value of the dependent variable when the independent variable is 3

3 $g(7) = -6$

4

a 2

b 0

c 1

5 46

6 -189

7

a 12

b -16

8

a 8

b -13

9

a $\frac{35}{4}$

b -7

10

a 1

b $-\frac{13}{5}$

11 $\frac{9}{2}$

12

a $a^2 + 8a$

b $b^2 + 8b$

13

a 12

b $m^2 + 8$

14

a

i $f(x) = 2 - \frac{1}{3}x$

ii 2

b

i $f(x) = \frac{6}{5}x + \frac{7}{5}$

i 5

c

i $f(x) = 3 - x - 6x^2$

ii -54

15

a $f(x) = \frac{1}{4}x - 2$

b 1

16

a

i $f(x) = \frac{7 - 5x}{-3}$

ii 1

b

i $f(x) = 4x^2 + x + 5$

ii 23

17 -10

18 -69

19

a True

b False

20 $P(2) = 14\,000$, so after 2 years there are$14\,000$ fish in the lake.

Additional questions

21

a $W(19)$ represents the amount of money Sharon makes if she works 19 hours.

b The equation $W(n) = 165$ represents that Sharon makes $\$165$ after working n hours.

c $W(26) = \$390$

22

- **a** The equation $h(4) = 16$ represents that the tomato plant will be 16 inches tall after 4 weeks.
- **b** 64 inches

23 Since $f(4) = 7$, we know that $4a + b = 7$. Since $f(-1) = -3$, we know that $-a + b = -3$. Solving this system of equations gives us $a = 2$ and $b = -1$.

24 An example scenario could be: Maggie has set aside 5 jelly beans for herself and will share 120 with her friends. Let x represent the number of people sharing the jelly beans (Maggie included) and let $f(x)$ represent the total number of jelly beans that Maggie ends up with.